Chapter 14, Problem 2RQ

Problem

On the Bode magnitude plot, the slope of 1⁄(5 + *jω*)2 for large values of *ω* is

(a) 20 dB/decade

(b) 40 dB/decade

(c) −40 dB/decade

(d) −20 dB/decade

Step-by-step solution

Step 1 of 1

Consider the following factor $\frac{1}{(5+j\omega)^2}$

For the factor, corner frequency is at $\omega = 5$ rad/s. for the pole with corner frequency at $\omega = 5$, the slope of the magnitude plot is -40 dB/decade due to power of two.

Therefore on the Bode magnitude plot, the slope of $\frac{1}{(5+j\omega)^2}$ for large values of ω is -40 dB/decade . Thus correct option is $\boxed{(c)}$.

20 dB/decade is for the zero. So option (a) is wrong.

40 dB/decade is for the zero due to power of two. So option (b) is wrong.

-20 dB/decade is for the pole. So option (d) is wrong.